

[Aim: 100/100 in Maths]

आरंभ

REAL NUMBERS

Lecture #4

ATK²:-

→ HCF & LCM by prime factorisation.

→ Prime & Composite No.

⊗ for two prime no. (p & q)

HCF = 1

LCM = $p \times q$

→ Completely divide ~~हरदहर~~

$R = 0$

Ye **Factor** aur **Multiple** Kya Hota Hai ?

FACTOR ? → जो divide कर दें।



6 के factor

→ जो 6 को divide कर दें



MULTIPLE ÷ जो table में आ जाए / जो divide हो जाए

Q is 9 a multiple of 3? (Yes)
क्या '9' → 3 की table में आता है?

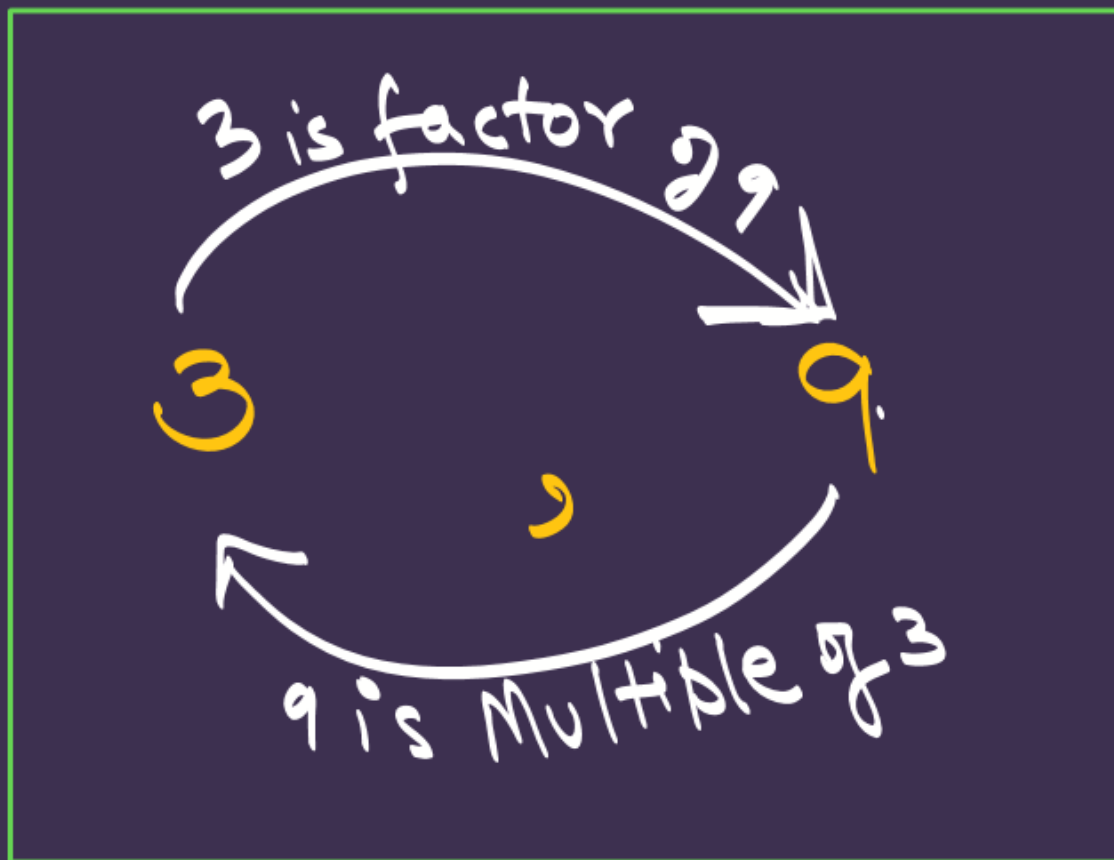
or, क्या 3 → 9 को divide कर देगा?

A is 14 is multiple of 2? (Yes)

Q is 97 a multiple of 9? (Yes)

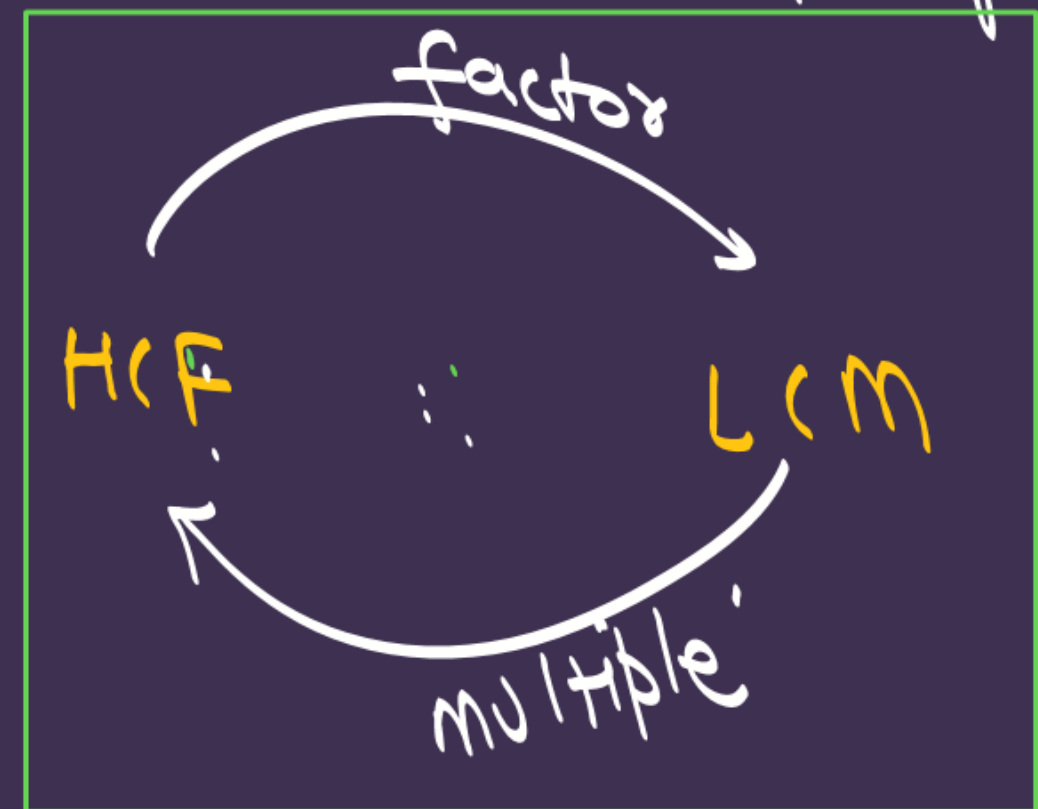
3B

Factor $\rightleftharpoons$ Multiple



3B

- ★ HCF is factor of LCM.
- ★ LCM is multiple of HCF



#LP: Assertion (A): For two odd prime numbers x and y ($x \neq y$),
 $\text{LCM}(2x, 4y) = 4xy$ True
Reason (R): $\text{LCM}(x, y)$ is a multiple of $\text{HCF}(x, y)$. T [CBSE 2025]

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). 52.59.1.
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). 26.82.1.
- (C) Assertion (A) is true, but Reason (R) is false. 6.12.1.
- (D) Assertion (A) is false, but Reason (R) is true. 14.47.1.

let $x = 3, y = 5$
 $\text{LCM}(6, 20) = 60$

$4xy \Rightarrow 4 \times 3 \times 5$
 $\Rightarrow 60$

#LP: If 'p' and 'q' are natural numbers and 'p' is a multiple of 'q', then what is the HCF of 'p' and 'q'?

(A) pq 25.57%

(B) p 29.39%

☒ (C) q 40.74%

(D) p + q 4.31%

ex: let p is multiple of q.
4 is multiple of 2.

$$\begin{aligned} 4 &= 2 \times 2 \\ 2 &= 2 \\ \text{HCF} &= 2 \end{aligned}$$

$$\begin{aligned} p &= 4 \\ q &= 2 \\ \text{HCF}(4, 2) &= 2 = q \end{aligned}$$

असम - असम

#LP: Which of the following statements is true for HCF and LCM of two distinct natural numbers a and b?

(i) HCF is always greater than LCM. (F) det 2 & 3

(ii) HCF is a factor of LCM. (T)

(iii) LCM is a factor of HCF. (F)

multiple

[CBSE 2026]

A. (i) only

B. (i) and (iii)

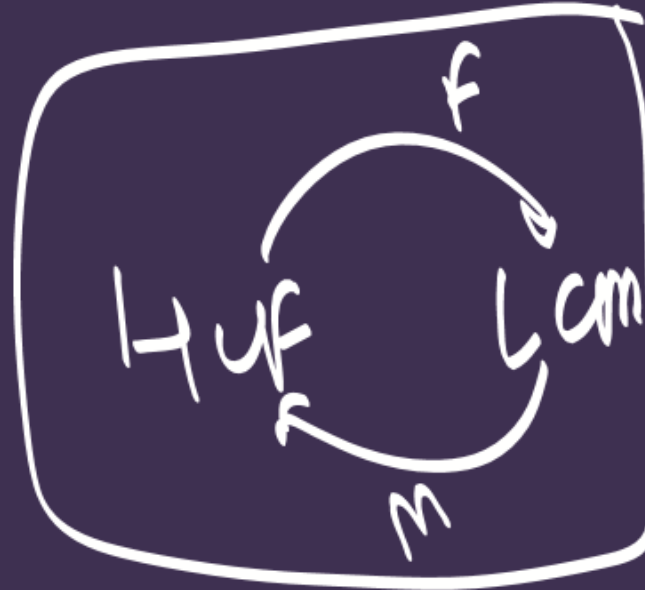
C. (i) and (ii)

D. (ii) only ✓

सं. 921

$$\text{HCF} = 1$$

$$\text{LCM} = 2 \times 3 \Rightarrow 6$$



#LP: Let n be a natural number. Then, the LCM ($n, n + 1$) is:

(A) $n \rightarrow 2$

(B) $n + 1 \rightarrow 3$

☒ (C) $n(n + 1) \rightarrow 2(3) = 6$

(D) 1

let $n = 2$

$\text{LCM}(2, 3)$
↓
prime

$\text{LCM} = 2 \times 3$
 $\text{LCM} = 6$

#LP: Assertion (A) : H.C.F. ($36m^2$, $18m$) = $18m$, where m is a prime number. True

Reason (R) : H.C.F. of two numbers is always less than or equal to the smaller number. True [CBSE 2026]

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
☒ (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

$$\text{HCF} \leq \text{smaller number}$$

eg

let two no's are 3 & 9. small

$$\begin{aligned} 3 &= 3 \\ 9 &= 3 \times 3 \end{aligned}$$
$$\text{HCF} = 3$$

$$\begin{aligned} 36 \text{ m}^2 &= 2 \times 2 \times 3 \times 3 \times m \times m \\ 18 \text{ m} &= 2 \times 3 \times 3 \times m \end{aligned}$$

$$\text{HCF} = 2 \times 3 \times 3 \times m$$

$$\boxed{\text{HCF} = 18m}$$

$$\begin{array}{r|l} 2 & 36 \\ \hline & 18 \\ \hline & 9 \\ \hline 3 & \\ \hline & 3 \\ \hline 3 & \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 18 \\ \hline & 9 \\ \hline & 3 \\ \hline 3 & \\ \hline & 1 \end{array}$$

5. The greatest number which when divides 1251, 9377 and 15628 leaves remainder 1, 2 and 3 respectively is

(a) 575

(b) 450

(c) 750

(d) 625

Part - 2 (Basic Subjective Type Questions)

3. $P = 2(4)(6)\dots\dots\dots(20)$ and $Q = 1(3)(5)\dots\dots\dots(19)$. What is the HCF of P and Q.



Exercise-2

Part - 1 (Advanced Multiple Choice Questions)

6. Let a and b be two positive integers such that $a = p^3q^4$ and $b = p^2q^3$, where p and q are prime numbers. If $\text{HCF}(a, b) = p^m q^n$ and $\text{LCM}(a, b) = p^r q^s$, then $(m + n)(r + s) =$
- (a) 15 (b) 30 (c) 35 (d) 72

7. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then the result obtained by dividing the product of the positive integers by the LCM (a, b) is :

(a) xy

(b) xy^2

(c) x^3y^3

(d) x^2y^2

Part - 2 (Advanced Subjective Type Questions)

1. Prime factorisation of three numbers A, B and C is given below :

$$A = (2^r \times 3^p \times 5^q)$$

$$B = (2^p \times 3^r \times 5^q)$$

$$C = (2^p \times 3^q \times 5^r)$$

such that , $p < q < r$ and $p, q,$ and r are natural numbers.

- The largest number that divides A, B and C without leaving a remainder is 30.
- The smallest number that leaves a remainder of 2 when divided by each of A, B and C is 5402

Find A, B and C. Show your work.

4. P is the LCM of 2, 4, 6, 8, 10; Q is the LCM of 1, 3, 5, 7, 9 and L is the LCM of P and Q. Then, find the relation between L and P.

5. On the two real numbers $a = 2 + \sqrt{5}$ and $b = 3 - \sqrt{7}$, perform the following operations :

[CBSE CFPQ]

- (i) Calculate the sum $(a + b)$
- (ii) Calculate the product (ab)
- (iii) Find the additive inverse of a
- (iv) Rationalise $1/b$
- (v) Verify whether the numbers a and b are rational or irrational. Provide a valid reason for your answer

Part - 3 (Advanced Miscellaneous Questions)

6. Match the Column I with Column II and choose the correct option :

Column-I	Column-II
(A) The HCF of the smallest number of two digits and the largest multiple of 5 which is less than 40 is	(i) 5
(B) HCF of $x^3y^4z^2$ and $x^2y^3z^5$, where x, y and z are distinct prime numbers is	(ii) 4
(C) The total number of factors of a prime number is	(iii) 2
(D) The sum of exponents of prime factors in the prime factorisation of 196 is	(iv) $x^2y^3z^2$

- (a) A-iv, B-ii, C-i, D-iii
- (b) A-i, B-iv, C-iii, D-ii
- (c) A-ii, B-iv, C-i, D-iii
- (d) A-ii, B-iv, C-iii, D-i